Gray Reid

Computational Modeling and Machine Learning, NYC

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Skills

Scientific Computing

- HPC, Numerical Methods, Computational Modeling
- Finite Difference, Finite Element, Spectral Methods, Monte Carlo, AMR, Multigrid
- Optimization, Profiling, Debugging

Machine Learning

- Neural Networks, Deep Learning, Reinforcement Learning
- Attention Mechanisms, Transformers, CNNs, RNNs, GANs, Diffusion Models, GRUs, LSTMs

Languages and Frameworks

- C, Python, Fortran
- Pytorch, Jax, TensorFlow
- MPI, OpenMP, CUDA

Numerical Analysis

- Error analysis, Stability Analysis, Complexity Analysis, Convergence Analysis
- Conditioning, Discretization

Mathematics and Physics

- Calculus, PDEs, SDEs, ODEs
- Lagrangian Mechanics, Hamiltonian Mechanics
- Quantum Mechanics, Statistical Mechanics, Thermodynamics
- Field Theory, Fluid Dynamics, Electromagnetism, General Relativity
- Linear Algebra, Probability, Statistics

Research

- Research Design, Scientific Writing, Literature Review
- Data Analysis, Data Visualization
- Mentoring, Curriculum Development, Presentations

Education

PhD Physics
Vancouver, BC, Canada

University of British Columbia — *2024*

- Developed high-performance numerical simulations of gravitational collapse improving resolution and accuracy by several orders of magnitude over state-of-the-art AMR simulations without significantly increasing computational cost
- Developed expertise in computational modeling, numerical methods, high performance computing, debugging and optimization of massively parallel systems

BSc Honours Physics
Wolfville, NS, Canada

Acadia University — *2012*

- Highlighted deficiencies in existing techniques for modeling TEM imaging of materials and developed new methods for addressing these deficiencies
- Developed improved equations describing cavity ringdown spectroscopy for use in chemical sensing applications

Profile

- Computational physicist with a PhD from UBC, combining expertise in mathematical modeling, data analysis, and machine learning. Eager to apply advanced quantitative skills and algorithm development experience to solve complex problems in quantitative finance and trading strategies.
- Seeking to transition from academic research to projects in AI or quantitative finance. Eager to apply advanced mathematical modeling and machine learning techniques to tackle challenging problems in industry. Committed to leveraging strong research and teaching background to drive innovation and contribute to high-impact teams in tech or finance sectors.

Select Publications

Universality in the Critical Collapse of the Einstein-Maxwell System

Physical Review D — *2023*

- Improved resolution and accuracy of in-lab adaptive mesh refinement (AMR) software for simulation of generic PDEs
- Improved resolution and accuracy by several orders of magnitude over previous investigations of the Einstein Maxwell system
- Resolved conflicts with previous investigations and provided a framework for understanding discrepancies in related literature.

Reference Metric Approach to the Z4 System

Physical Review D — *2023*

- Derived a novel hyperbolic formulation of General Relativity using a reference metric approach
- Achieved performance equivalent or exceeding leading formulations of General Relativity such as GBSSN and FCCZ4
- Demonstrated the stability of the formulation through a pseudodifferential first order reduction and simulations in the strong field regime

Stability of Nonminimally Coupled Topological-Defect Boson Stars

Physical Review D — *2024*

- Performed both fully non-linear and semi-analytic stability analysis for stationary solutions in f(R) gravity
- Resolved the open question of stability for topological-defect boson stars in f(R) gravity

Select Awards

NSERC CGSD
University of British Columbia

Four Year Fellowship
University of British Columbia

NSERC CGSM
University of British Columbia

Governor General's Medal
Acadia University